Project 30: Euclidifying Gaia

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Project summary

The goal of this project is to use machine learning models to classify objects in the Gaia DR3 catalogue, focusing on non-point-like objects. Firstly, we would like to identify them as such, and secondly we want to be able to apply labels such as “galaxy nucleus”, “binary star”, “star-forming region”, etc. As the full-frame images from Gaia are not downloaded to ground, the data we have is a collection of different statistics on each Gaia source provided in the DR3 catalogue. However, for a subset of the Gaia data we have Euclid catalogues (and images) from which we can establish which of the Gaia catalogue statistics are useful in identifying and labelling non-point-like objects.

Background

Gaia and Euclid are two ESA flagship space observatories, equipped with cameras of similar resolution. Euclid sends its enormous (~100 times larger than HST’s) images to the ground, but Gaia does not. This is because Gaia’s field of view is even larger than Euclid’s but the image depth is shallower, so quite often, most of the image is dark, and hence not worth the ground station time to download. Instead, objects on the Gaia focal plane, such as stars, galaxies, quasars, asteroids, are detected on board and only small, compressed cut-outs are sent to Earth. As Gaia’s main goal is to measure properties of stars, the on-board detection algorithm is optimised for point-sources, however no algorithm is perfect, and many non-stellar objects are detected. These include for example, the nuclei of galaxies, bits of galaxies ’spiral arms, chunks of planetary nebulae etc.

Gaia continuously scans the sky, coming back to the same spot many times as it builds up its full-sky catalogue. The catalogue statistics we are interested in are derived from these multiple observations of each source, as these will likely be different for point-like and extended sources. These differences arise from Gaia’s increased resolution in the along-scan direction over that of the across-scan direction. This means that for extended objects their orientation with respect to Gaia’s scan direction will affect their measurements and even whether they are detected on that pass at all.

Although most diffuse structures on the sky are not recorded by Gaia, many high surface brightness knots and blobs are. The goal of this project is to train a machine learning model to reconstruct the origin and the context of the non-stellar objects measured by Gaia using their properties in the published catalogue with the help of high-resolution Euclid images.

Project details

You will need to cross-match Euclid (ERO, Q1) and Gaia DR3 catalogues to build the training and the validation data-sets.

The simplest way to start to understand which Gaia catalogue statistics are helpful in identifying non-point like sources may be by using the XGBoost package, which provides a simple and efficient means of carrying out supervised learning. To this end, it will be beneficial to familiarise yourself with the contents of the Gaia DR3 catalogue and understand how the catalogue statistics are generated. You will need to choose which catalogue statistics from the Gaia catalogue to use as inputs.

Once it is understood how to find non-point like sources in Gaia, the Euclid images can be used to see if these sources can be further classified. What information about non-stellar objects can we actually extract? Will it be just the crude classes such as “galaxy nucleus”, “binary star”, “star forming region” or can we provide more useful details on the origin and properties of the source, as well as its surroundings?

A large variety of ML methods are currently available, and your task will be to identify the most suitable ones to achieve the goals of the project. To deal with the missing data and/or data with large error-bars in the Gaia statistics gracefully, and to provide uncertainties for the classification, we might investigate building a model based on Gaussian processes. It may also be useful to consider unsupervised techniques, such as AstroCLIP (Parker et al 2023).

Skills required

The project will be computationally based; knowledge of Python is highly desirable together with knowledge of /or willingness to learn how to query a PostgreSQL database. Support will be provided.

Useful References (List of important papers/review articles relevant to the project)

- Cuillandre et.al. https://arxiv.org/abs/2405.13496
- Euclid Q1 data release https://www.cosmos.esa.int/en/web/euclid/euclid-q1-data-release
- The Gaia Archive https://gea.esac.esa.int/archive/
- Gaia DR3 documentation https://gea.esac.esa.int/archive/documentation/GDR3/index.html
- AstroCLIP https://arxiv.org/abs/2310.03024
- XGBoost https://xgboost.readthedocs.io/en/stable/index.html
- UMAP https://umap-learn.readthedocs.io/en/latest/

General References (List of papers referred to in the project)

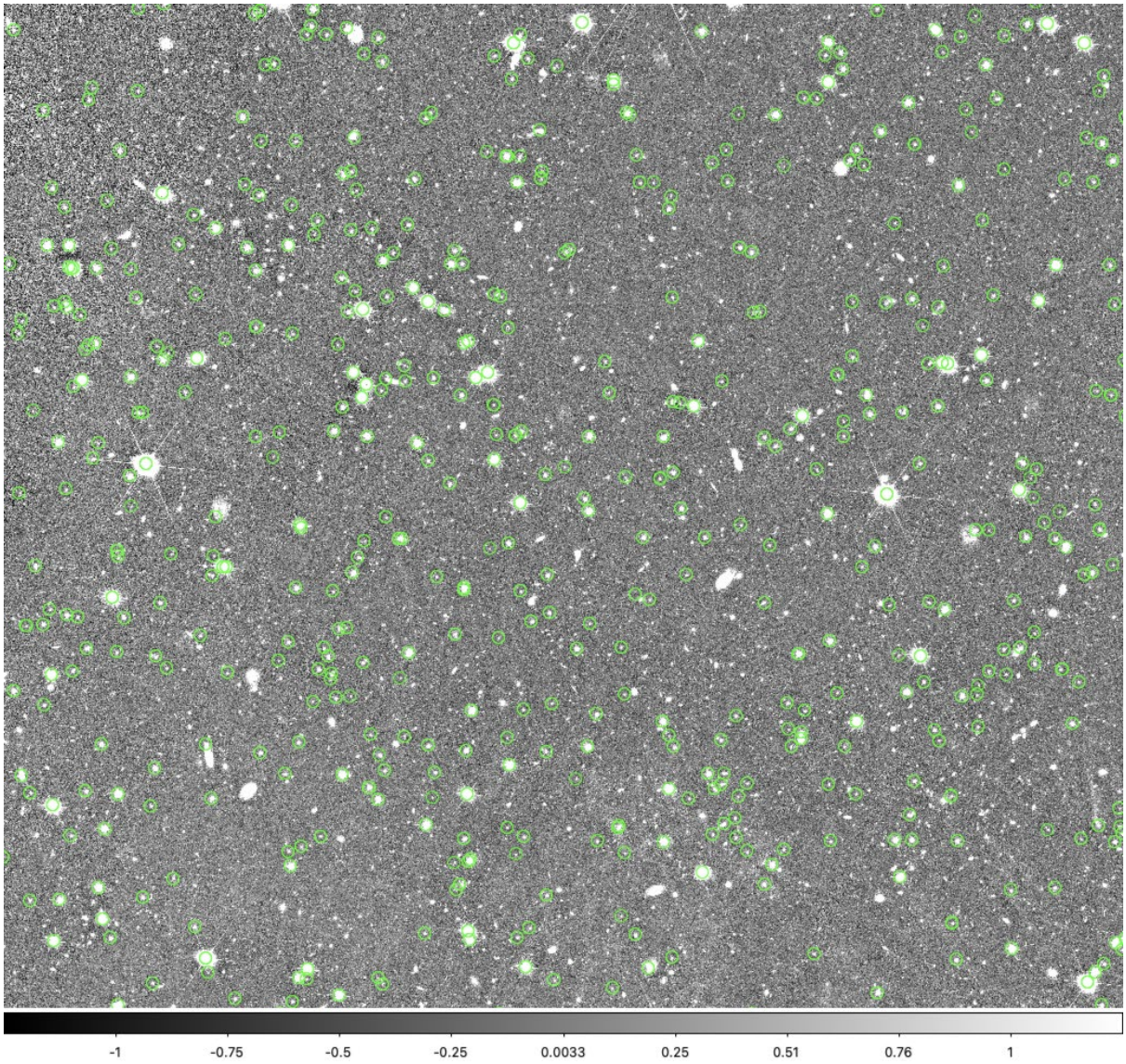


Figure 1: Gaia DR3 source positions overlaid on a small part of the Euclid Image of Abell 2390. The greater depth of the Euclid image is clearly apparent, and if you look carefully you can find Gaia detections connected with extended emission.